

ARTICLE

Movement Patterns of Juvenile Sablefish within a Nursery Area in Southeast Alaska

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Abstract

The Sablefish *Anoplopoma fimbria* is a commercially important species that inhabits waters of the Pacific Ocean from Baja California, Mexico, to the Bering Sea, Alaska. Most studies have focused on the larval, neustonic (near-surface) juvenile, or adult stage of the Sablefish life cycle, while much less is known about the postsettlement juvenile stage (ages 0–2) in nearshore nursery areas, even though survival of postsettlement juveniles is thought to be important for determining year-class strength. We used acoustic telemetry to monitor movement of postsettlement juvenile Sablefish in a nursery area in Alaska to better describe their period of nearshore residency. Forty juvenile Sablefish received surgically implanted acoustic transmitters and were monitored using eight fixed receivers throughout the summer and fall of 2015 and 2016 in a small bay (St. John Baptist Bay) in Southeast Alaska. We quantified movement patterns in terms of displacement from the head of the bay, distance traveled, and duration of time spent within the bay for 28 individuals. Sablefish showed fidelity to the bay during the summer, with relatively high rates of movement within the bay. Juvenile Sablefish traveled 9.4 km/d and spent 20.2 h/d within the bay on average in 2015 and traveled 13.0 km/d and spent 17.9 h/d within the bay on average in 2016. Tagged Sablefish showed the greatest affinity for a region near the head of the bay, perhaps indicating an area of preferred habitat, prey resources, or environmental conditions. In addition, we assessed variation in horizontal movements among individuals and identified three distinct movement types. This study fills a gap in knowledge of Sablefish early life history by characterizing movement during nearshore residency before out-migration into deeper waters. Our results

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reinforce the importance of nearshore habitats like St. John Baptist Bay for juvenile Sablefish prior to recruitment into fisheries.

Estuarine and coastal ecosystems have long been recognized as nursery grounds for crustaceans and fishes (Herke 1971; Weinstein 1979; Boesch and Turner 1984; Beck et al. 2001). Nursery habitats support ecological processes that contribute to recruitment success by improving growth and survival of juveniles, such as providing refuge habitat from predators, food resources, and favorable environmental conditions at a range of spatial and temporal scales (Heck et al. 2003; Nagelkerken et al. 2015; Sheaves et al. 2015). Assessing how juvenile fish use nursery areas may contribute to an understanding of the connection between postsettlement habitat and the population's vital rates (e.g., survival, recruitment, and emigration; Rutecki and Varosi 1997a; Hanselman et al. 2015; Pirtle et al., in press). One way to assess how juvenile fish use nursery areas is to examine their movements within such habitats. Fish movements within nearshore habitats may be related to ecological factors, such as intraspecific competition for territories, foraging strategies, predator avoidance, and seasonal migrations (Boesch and Turner 1984; Roy et al. 2013; Finn et al. 2014; Henderson et al. 2014; Sheaves et al. 2015). Various methods can be used to investigate the movement of fishes, including distributional studies that compare fish abundance and size structure among areas; conventional tagging studies that allow for greater spatial and temporal resolution; and electronic tagging and acoustic telemetry studies that provide more detailed information on movements (Furey et al. 2013).

In this study, we examined the movement patterns of postsettlement juvenile Sablefish *Anoplopoma fimbria*, a commercially important species in the North Pacific, within a nearshore nursery habitat. Adult Sablefish inhabit waters of the Pacific Ocean along the continental slope at depths of 200 to 1,000 m, with a geographic distribution ranging from Baja California, Mexico, to the Bering Sea, Alaska, and west to Japan (Sasaki 1985; Wolotira et al. 1993). Mark-recapture studies suggest that there are southern and northern West Coast Sablefish populations, with some mixing near Vancouver Island (McDevitt 1990; Kimura et al. 1998). Sablefish of the northern population spawn offshore from February to March at depths up to 500 m (Mason et al. 1983; McFarlane and Nagata 1988; Wing 1997). After the eggs incubate and hatch, the larvae rise to the surface, reaching nearshore nursery habitats by the end of their first summer, where they transition to demersal habitats (Mason et al. 1983; Kendall and Matarese 1987; Rutecki and Varosi 1997a; Wing 1997). Postsettlement juvenile Sablefish remain in nearshore waters for

1–2 years, reaching approximately 300 to 400 mm in length before they emigrate to deeper waters of the continental shelf, eventually arriving at adult habitats of the continental slope by age 4–5 (Mason et al. 1983; Kendall and Matarese 1987; Rutecki and Varosi 1997a; Maloney and Sigler 2008). Bays and inlets from Southeast Alaska to British Columbia are thought to be important nursery grounds for juvenile Sablefish (Sasaki 1985; Rutecki and Varosi 1997a, 1997b).

Relatively little is known about juvenile Sablefish ecology in nearshore areas; however, it is during this postsettlement period when factors influencing Sablefish body condition and energy allocation strategy are thought to play an important role in determining year-class strength (Mason et al. 1983; Sigler et al. 2001, 2003). Juvenile Sablefish growth rates are among the highest measured in juvenile teleosts; young-of-the-year Sablefish can grow at a rate of over 3 mm/d under optimal conditions (Sogard and Olla 2001). While residing in nearshore habitats, juvenile Sablefish are opportunistic feeders, eating high-energy, seasonally pulsed prey, including Pacific Herring *Clupea pallasii* and tissue from spawning adult Pacific salmon *Oncorhynchus* spp. (Coutré et al. 2015). Their vertical movements may be related to foraging, as suggested by a study of six juvenile Sablefish tagged in a Southeast Alaska bay, which found that individuals spent most of their time near the bottom while embarking on periodic vertical excursions, primarily during dawn and day periods (Coutré et al. 2017). Beyond these studies of juvenile Sablefish feeding ecology and vertical distribution (Courtney and Rutecki 2011; Coutré et al. 2015, 2017), little is known about the occupancy of postsettlement juvenile Sablefish within nearshore areas over multiple seasons or their degree of movement within nearshore habitats prior to out-migration.

We used acoustic telemetry to examine movement patterns by juvenile Sablefish in a nursery habitat during their postsettlement stage. The first objective of the study was to describe movement of juvenile Sablefish throughout St. John Baptist Bay, Alaska, a known nursery area (Rutecki and Varosi 1997a). We hypothesized that juvenile Sablefish would move throughout the entire bay but would show stronger fidelity to the head of the bay (1) due to its proximity to freshwater input, which offers a potential food source of salmon carcasses during the late summer and fall; and (2) because it has been found to produce higher catch rates of juvenile Sablefish (Rutecki and Varosi 1997a; Courtney and Rutecki 2011; Coutré et al. 2015,

2017). The second study objective was to characterize differences in movement patterns among individual Sablefish, as individual variation in movement has been documented for other fishes and may be driven by dynamic ecological factors like foraging strategies, avoiding predators, and biological responses to environmental conditions (Boesch and Turner 1984; Beaudreau and Essington 2011; Roy et al. 2013; Finn et al. 2014; Henderson et al. 2014; Sheaves et al. 2015). We hypothesized that individual Sablefish would exhibit variation in their degree of movement and location within the bay, with some individuals showing fewer absences between detections and stronger site fidelity than others. We also expected that larger

individuals, which may be closer to the age of out-migration, would show higher movement rates and lower residence times in the bay compared to smaller individuals.

METHODS

Study area.—The study was conducted from June 2015 to October 2016 in St. John Baptist Bay (SJBB), Baranof Island, approximately 33 km north of Sitka, Alaska. St. John Baptist Bay is a small bay (about 3 km long and <1 km wide) that opens to Salisbury Sound, has a freshwater input, and was historically a commercial log transfer facility (Miller et al. 2007; Figure 1). Depth ranges

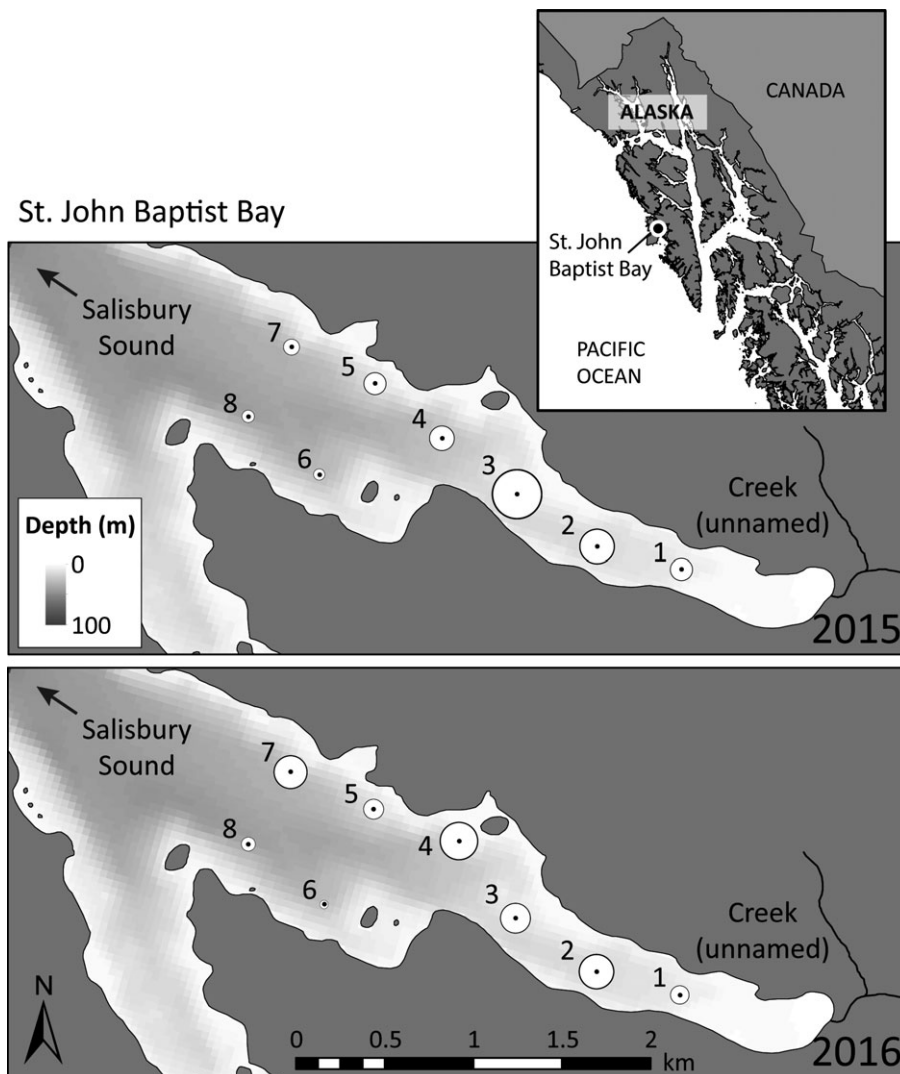


FIGURE 1. Map of St. John Baptist Bay, Alaska, showing receiver placement (solid points) and mean residence index (RI; open circles) for 2015 (top) and 2016 (bottom). Open circles are proportional to the mean RI across all fish for each receiver in each study year (see Table 2 for mean RI values). The bay has a freshwater source at the head (unnamed creek), and the mouth opens into Neva Strait, toward Salisbury Sound. A 40-m raster data set showing depth in the bay was provided by the National Oceanic and Atmospheric Administration and the Nature Conservancy (source: http://seakgis.alaska.edu/data/bathy_40m.zip).

from 60 to 80 m at the mouth of the bay and becomes gradually shallower toward the middle-head region of the bay, where the depth ranges from 20 to 30 m until nearing the freshwater source at the very head of the bay, at which the depth quickly shallows to less than 15 m. Bottom substrate is mostly soft sediment throughout, with the southern side of the bay near the mouth exhibiting rocky features and boulders. We selected SJBB as the study location for this research because it has been the focus of several juvenile Sablefish ecological studies across multiple seasons and years (e.g., Rutecki and Varosi 1997a; Maloney and Sigler 2008; Courtney and Rutecki 2011; Coutr  et al. 2015, 2017) and continues to be sampled annually during the National Marine Fisheries Service (NMFS) juvenile Sablefish tagging survey (Dana Hanselman, National Oceanic and Atmospheric Administration [NOAA], personal communication).

Tagging.— We monitored juvenile Sablefish movements using acoustic telemetry. Acoustic telemetry is a technique in which fish tagged with acoustic transmitters are detected by submerged hydrophones to provide detailed observations of distribution and movements (Eiler and Bishop 2016). Twenty juvenile Sablefish were captured for tagging on June 10, 2015, and twenty were captured on June 11 and 12, 2016. Sample size was limited to 20 fish each year based on the size of the bay, potential tag interference, and boat traffic within the bay (Jonathan Mulock, Vemco, personal communication). Based on length ranges reported in the literature, tagged juvenile Sablefish were likely age-1 (280–350 mm) and age-2 (390–510 mm) fish (Rutecki and Varosi 1997a). Sablefish were caught using rod and reel with baited hooks near the head of the bay at depths ranging from 11.3 to 25.9 m. Date, time, geographic position, and depth were recorded for each fish upon capture. Fish were individually sedated using 10% eugenol (Aqui-S 20E) at a concentration of approximately 30 mg of Aqui-S per liter of seawater (average temperature = 10.3°C; measured to the nearest 0.001°C) and were weighed (round mass, nearest 10 g) and measured (FL, nearest 5 mm) before tagging.

Acoustic transmitters (Vemco V9-2L 69 kHz; 146 dB referenced to 1 μ Pa at 1 m; 30–90-s delay; 9-mm diameter $\times$ 29-mm length; weighing 4.7 g in air and 2.9 g in water) were inserted through an approximately 1-cm incision made through the midline of the ventral musculature of each juvenile Sablefish. The incision was closed by two simple interrupted sutures in accordance with procedures outlined by Lowe et al. (2003) and Mulcahy (2003). All instruments and tags were thoroughly disinfected in a 10% betadine solution between surgeries. When transmitters are less than 2% of the weight of the fish, adverse effects of the tagging procedure are thought to be minimal (Gallepp and Magnuson 1972; Ross and McCormick 1981; Winter 1996); on average, transmitters were 1.4% of the

initial weight of individual tagged Sablefish. Each fish was also tagged externally with a unique T-bar anchor tag (Floy) to permit visual identification if fish were recaptured. After surgery, each fish was immediately placed in a recovery tank of fresh seawater for observation until it was apparent that the fish had regained equilibrium and a normal gill ventilation rate. On average, sedation and surgery took fewer than 10 min total for each fish, with post-surgical recovery time ranging from approximately 10 to 35 min. Fish that did not behave normally or did not appear to be in healthy condition after an extended recovery period were sacrificed; all other fish were released at their approximate capture location near the head of the bay. The research was approved by the Institutional Animal Care and Use Committee at the University of Alaska Fairbanks (Protocol Number 738825). Transmitters implanted during 2015 were set to expire after 370 d to avoid inference with the transmitters deployed the following year; the lifetime for transmitters implanted in 2016 was estimated to be 476 d.

Fixed receiver array and mobile surveys.— Between June 9, 2015, and October 11, 2016, eight Vemco VR2W acoustic receivers were deployed and maintained in SJBB. Each acoustic receiver station was submerged and anchored using a 23-kg cement pier block, and receivers were suspended approximately 3 m off the bottom beneath a 20-cm trawl float on polypropylene line. The receivers were attached hydrophone-down below the trawl float to more effectively capture detections of the juvenile Sablefish, as they typically inhabit the area near the benthos. The receivers were positioned to provide the greatest possible coverage of the study area and remained in the bay throughout the duration of the study, except from March 25 to April 10, 2016, during the opening of a commercial herring sac roe fishery. Receivers 7 and 8 were removed from May 17 to June 13, 2016, due to entanglements with commercial troll gear. These gaps in coverage did not affect our results, as they occurred after fish tagged in 2015 were no longer detected in the bay and before fish were tagged in 2016.

The acoustic array was designed to monitor fish movement throughout the entire bay and to create an acoustic gate to capture movement in and out of the bay. To help determine the siting of the receiver array within SJBB, range testing was performed on receivers positioned at three different points between the head and mouth of the bay (approximately at receivers 1, 5, and 6; Figure 1) to ensure that receiver detection capabilities did not vary by location in the bay. For each receiver location, we drove transects ($n = 6$) from the receiver position to approximately 1,000 m away with a tag suspended from the boat. The geographic position of each transmission was recorded using the boat-based hydrophone equipped with GPS (Vemco VR100). Using a comparison of transmissions detected by the boat-based hydrophone to

transmissions detected by the stationary receiver, we modeled the probability of detection for each receiver location using logistic regression. Range testing results showed a 50% probability of detection at distances from 295 to 695 m (mean = 440 m) from stationary receivers, depending on the position of the receiver and local conditions during range testing. Based on these detection capabilities, the receiver array provided nearly full coverage of the bay. One reference tag was deployed each year at the mouth of the bay between receivers 7 and 8 to test the functional performance of the receiver array, even when tagged fish were not present.

In addition to fixed receivers, individual fish were detected from a boat by using the boat-based hydrophone once every 7–10 d from June through August of each year to ground truth the presence/absence of tags in the bay against receiver data and to allow for movement observations in season.

Data analysis.— We examined the horizontal movement of juvenile Sablefish throughout SJBB and characterized variation in movement patterns among individuals. Prior to analysis, the first 5 d of each study year were excluded to account for the acclimation of tagged fish. Tags that were assumed to have been expelled and tags in fish that were presumed to have died (i.e., the tag did not change position for 30 d or longer) were also excluded. The data were filtered to remove false detections by defining presence within the detection array for a given day as being detected at least twice on any receiver in that 24-h period; absence from the detection array was defined as being detected only once or not at all in that 24-h period. Next, we assigned the primary receiver location of each tagged fish to every 15-min interval in the time series. Primary receiver location was determined as the receiver by which the fish was most frequently detected during that time interval; if more than one receiver was designated as primary for a given interval, a single receiver was randomly chosen from among those with the most detections. We used this method of assigning a primary receiver to each 15-min interval because for any given transmission, a fish could have been detected at more than one receiver due to overlap in receiver range. Although this approach does not generate a precise estimate of fish location, it provides a means of examining temporal patterns of movement across coarse regions of the bay.

Using primary receiver locations, we calculated residence indices (RIs) for each receiver and fish. The RI was calculated as the number of intervals during which an individual fish was detected at each receiver station divided by the total number of intervals in which the fish was detected anywhere within the acoustic array, similar to the method of Kessel et al. (2016) except that 15-min intervals were used instead of days. We mapped the mean RI across fish for each receiver and each year.

To examine Sablefish movement throughout SJBB, we calculated three metrics for each fish: (1) daily displacement from the receiver in closest proximity to the head of the bay (receiver 1); (2) daily distance traveled within the receiver array; and (3) daily duration of presence within or absence from the receiver array. Displacement was calculated as the straight-line distance (km) between the receiver on which a fish was detected and receiver 1 for each 15-min interval. This measurement allowed for visualization of movement from or toward the head of the bay over time. Daily displacement was calculated for each fish as the average displacement distance per 24-h period from the sixth day posttagging to the last date of detection and then was averaged by date for all fish. To calculate distance traveled and duration of time spent within the bay for each fish per 24-h period, a unique number was assigned to each movement event. When a fish moved to a new receiver or out of the detection array, this marked a new event comprised of distance traveled to that receiver and the duration spent at that receiver. For each movement event, distance was calculated as the straight-line distance between the fish's previous primary receiver location and the new primary receiver location, and duration was the sum of 15-min intervals for that movement event. Once the fish changed primary receiver locations, a new event began, and the distance and duration were calculated for that movement event. Daily distance traveled (km/d) within the receiver array was calculated by summing distance across events for each 24-h period from the sixth day posttagging to the last date of detection for each fish. Daily duration (h/d) was calculated by summing the duration across events for each 24-h period from the sixth day posttagging to the last date of detection for each fish. Box plots were created to show the median and range of estimated daily distance traveled and daily duration spent within the bay for individuals.

Because it was not possible to determine a tagged individual's precise location at any given time or movement path among receivers, these metrics served as coarse-scale indices of relative movement that allowed for an examination of variation in the degree of movement among individuals. It is important to note that displacement and distance measurements account for travel in straight-line distances using only detections inside the range of the receiver array; thus, they may overestimate or underestimate movement depending on the actual location of the fish. Direct distances between two receivers were generated using package "VTrack" in R (Campbell et al. 2012; R Core Team 2017).

We used two complementary approaches to characterize variation in movement patterns among individuals. First, we visually categorized each fish's movement based on the overall shape of its time series of displacement distance from receiver 1. The three unique groups were identified

and defined as follows: (1) fish with a greater proportion of detection intervals near the mouth of the bay; (2) fish with a greater proportion of detection intervals near the head of the bay; and (3) fish that were intermediate to these two types, exhibiting a more even distribution of detections between head and mouth (i.e., demonstrating more dispersive tendencies). To corroborate this qualitative assessment, we calculated the proportion of 15-min detection intervals that fell into each of four regions of the bay: head (receivers 1 and 2), middle head (receivers 3 and 4), middle mouth (receivers 5 and 6), and mouth (receivers 7 and 8). We then used a *k*-means cluster analysis ($k = 3$; R Core Team 2017) to quantitatively categorize each fish into one of three groups based on similarities in their proportional frequency distributions across regions.

To evaluate the influence of body size on movement, we used separate linear regression models to test for relationships between FL and days present within the bay, FL and mean daily distance traveled, and FL and mean daily duration. All analyses were conducted in R (R Core Team 2017).

RESULTS

Summary of Tagged Individuals

Overall, 40 juvenile Sablefish were implanted with acoustic transmitters in 2015 and 2016, and 28 of these fish were used for analyses. The remaining 12 fish were eliminated from analyses due to suspected tag loss or mortality, as no movement was detected for 30 d or longer (i.e., until the battery expired or the study ended) for each omitted tag. Fork lengths ranged from 300 to 420 mm for both study years (mean $\pm$ SD = 349 ± 24 mm), and weights ranged from 240 to 690 g (352 ± 89 g; Table 1). Based on FLs of tagged fish, we assumed that the tagged individuals were predominantly age 1, though some may have been age 2 (Rutecki and Varosi 1997a). Between June 11, 2015, and October 11, 2016, a total of 3,886,245 detections were logged for all 40 tagged juvenile Sablefish. The 28 tagged fish used for analyses were detected in the receiver array for at least 5 d during the period of analyses (i.e., after the first 5 d posttagging were eliminated). The final data set used for analyses contained a total of 1,494,963 detections for 28 individuals; these detections were subset into 15-min intervals for a total of 106,793 detection intervals (Table 2). All eight receivers detected fish, and all fish were detected on every receiver except for one individual (identification number [ID] 43454) that was not detected again at receivers 1 and 2 after the initial 5-d acclimation period. Only one fish (ID 34759) tagged in 2015 and used in the analyses was detected again after October 21, 2015. It was detected 195 times between December 7 and 10, 2015; however, these detections were not included in this study.

Presence/Absence

Tagged Sablefish were present within the bay from 5 to 121 d (mean $\pm$ SD = 46 ± 39 d) between the date of first detection (i.e., after the 5-d acclimation period) and the date of last detection for each fish (Table 1). Absence from the bay ranged from 0 to 56 d (8 ± 12 d) over the same period. Individuals spent an average of 36 d in the receiver array in 2015, with 50% of tagged fish no longer detected within the bay by mid-July (Figure 2). Individuals spent an average of 61 d in the receiver array in 2016, with 50% of tagged fish no longer detected by early September (Figure 2). Mean daily absence for absences less than 24 h ranged from 0.5 to 4.1 h/d (mean $\pm$ SD = 1.4 ± 1.0 h/d) across individuals in 2015 and from 0.8 to 5.5 h/d (2.3 ± 1.4 h/d) in 2016. Some of this short-term absence could be due to being outside the detection radius of the receiver array or within acoustic shadows, although still inside the bay. Absences occurred throughout the summer and fall, with no obvious pattern among individuals (Figure 3). Receiver performance was generally stable over time, as the reference tags were detected throughout the study and there were minimal breaks between detections.

Residence Index

Mean RI values for tagged juvenile Sablefish in both years indicated an affinity to receivers 2 and 3 located in the middle-head region of the bay (Figure 1). Near the mouth of the bay, Sablefish favored the northern shoreline (receivers 5 and 7), whereas RI values for receivers 6 and 8 along the southern shoreline were much lower for both study years. Across both years, receivers 2 and 3 had high mean RI values, while receivers 6 and 8 had the lowest mean RI values. Receivers 4 and 7 also had high mean RI values for 2016.

Displacement, Distance, and Duration

The mean daily displacement for fish tagged in 2015 ranged from 0.3 to 2.6 km/d (mean $\pm$ SD = 1.5 ± 0.5 km/d across all fish), and the mean daily displacement for fish tagged in 2016 ranged from 0.7 to 2.3 km/d (1.6 ± 0.3 km/d across all fish; Figure 4). The daily distance traveled for fish tagged in 2015 ranged from 8.6 to 18.2 km/d (13.0 ± 2.6 km/d), and the daily distance traveled for fish tagged in 2016 ranged from 1.3 to 14.5 km/d (9.4 ± 4.1 km/d; Figure 5). The daily duration of time spent within the receiver array for fish tagged in 2015 ranged from 17.2 to 22.7 h/d (20.2 ± 1.6 h/d), and the daily duration of time spent within the array for fish tagged in 2016 ranged from 1.5 to 23.5 h/d (17.9 ± 5.8 h/d; Figure 6). Some fish moved more frequently throughout the bay, while others remained stationary for longer periods of time, as measured using distance and duration spent within the receiver array (Figures 5, 6).

TABLE 1. Summary information for individual tagged Sablefish in St. John Baptist Bay, Alaska (ID = identification number). “Study start” marks the sixth day posttagging, while “last detection” is the date of last detection or the end of the study. Range is the total number of days between the study start and last detection. “Association with bay, in (d)” represents the number of days each fish was detected in the detection array at least twice, while “association with bay, out (d)” represents the number of days each fish was not detected in the detection array at all or was detected only once. Presence rate was calculated by dividing the days in bay by the range.

Release			Monitoring				Association with bay		
Tag ID	FL (mm)	Weight (g)	Release date	Study start	Last detection	Range (d)	In (d)	Out (d)	Presence rate (%)
34754	350	450	Jun 10, 2015	Jun 15, 2015	Jun 26, 2015	12	12	0	100
34755	355	350	Jun 10, 2015	Jun 15, 2015	Jul 26, 2015	42	39	3	93
34756	350	340	Jun 10, 2015	Jun 15, 2015	—	—	—	—	—
34757	380	410	Jun 10, 2015	Jun 15, 2015	Aug 20, 2015	67	36	31	54
34758	345	300	Jun 10, 2015	Jun 15, 2015	Oct 20, 2015	128	121	7	95
34759	360	380	Jun 10, 2015	Jun 15, 2015	Sep 5, 2015	83	83	0	100
34761	370	400	Jun 10, 2015	Jun 15, 2015	Jun 21, 2015	7	6	1	86
34762	360	370	Jun 10, 2015	Jun 15, 2015	Jul 5, 2015	21	18	3	86
34763	360	360	Jun 10, 2015	Jun 15, 2015	Jun 26, 2015	12	12	0	100
34764	360	360	Jun 10, 2015	Jun 15, 2015	Jun 27, 2015	13	12	1	92
34765	350	330	Jun 10, 2015	Jun 15, 2015	—	—	—	—	—
34766	370	400	Jun 10, 2015	Jun 15, 2015	Jun 29, 2015	15	15	0	100
34767	340	300	Jun 10, 2015	Jun 15, 2015	Aug 31, 2015	78	78	0	100
34768	340	300	Jun 10, 2015	Jun 15, 2015	—	—	—	—	—
34769	380	420	Jun 10, 2015	Jun 15, 2015	Jul 18, 2015	34	28	6	82
34770	350	320	Jun 10, 2015	Jun 15, 2015	Jun 19, 2015	5	5	0	100
34771	340	320	Jun 10, 2015	Jun 15, 2015	Aug 4, 2015	51	44	7	86
34772	360	350	Jun 10, 2015	Jun 15, 2015	Aug 5, 2015	52	34	18	65
34773	360	350	Jun 10, 2015	Jun 15, 2015	Jul 11, 2015	27	27	0	100
34774	335	290	Jun 10, 2015	Jun 15, 2015	—	—	—	—	—
43438	325	280	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43439	345	310	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43440	400	580	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43441	365	420	Jun 12, 2016	Jun 17, 2016	Oct 5, 2016	111	96	15	86
43442	365	370	Jun 11, 2016	Jun 17, 2016	Jul 10, 2016	24	21	3	88
43443	340	370	Jun 11, 2016	Jun 17, 2016	Sep 4, 2016	80	59	21	74
43444	360	420	Jun 11, 2016	Jun 17, 2016	Jul 3, 2016	17	17	0	100
43445	345	350	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43446	300	240	Jun 12, 2016	Jun 17, 2016	Oct 9, 2016	115	110	5	96
43447	420	690	Jun 12, 2016	Jun 17, 2016	Jul 8, 2016	22	10	12	45
43448	315	260	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43449	355	430	Jun 12, 2016	Jun 17, 2016	Oct 9, 2016	115	115	0	100
43450	335	280	Jun 12, 2016	Jun 17, 2016	Oct 3, 2016	109	53	56	49
43451	330	310	Jun 12, 2016	Jun 17, 2016	Jul 4, 2016	18	11	7	61
43452	365	390	Jun 12, 2016	Jun 17, 2016	Jul 12, 2016	26	24	2	92
43453	310	240	Jun 12, 2016	Jun 17, 2016	Oct 9, 2016	115	115	0	100
43454	330	320	Jun 12, 2016	Jun 17, 2016	Oct 4, 2016	110	97	13	88
43455	320	250	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43456	310	240	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—
43457	310	240	Jun 12, 2016	Jun 17, 2016	—	—	—	—	—

Movement Types

Movement varied among individual Sablefish, and three movement types were apparent from visual classification

of individual plots of displacement distance from receiver 1 over time (Figure 7). Ten fish increased distance from receiver 1 over time, making few excursions back to the

TABLE 2. Summary information for acoustic receivers placed in St. John Baptist Bay, Alaska, with bottom depth, displacement distance from receiver 1 (R1), total number of 15-min intervals, and mean residence index (RI) for each study year for the 28 Sablefish used in analyses.

Receiver	2015 receiver data				2016 receiver data			
	Depth (m)	Displacement from R1 (km)	Number of detection intervals	Mean RI	Depth (m)	Displacement from R1 (km)	Number of detection intervals	Mean RI
1	13.23	0.00	5,567	0.114	13.02	0.00	4,418	0.098
2	23.90	0.49	10,892	0.179	23.78	0.49	10,819	0.181
3	18.35	1.02	13,869	0.256	17.07	1.02	10,003	0.157
4	42.07	1.53	3,598	0.128	18.90	1.51	10,141	0.196
5	22.07	2.01	7,138	0.118	23.48	2.02	7,780	0.105
6	52.74	2.10	1,972	0.061	46.04	2.07	1,496	0.042
7	47.26	2.52	2,239	0.082	46.95	2.53	11,425	0.172
8	71.65	2.58	1,918	0.063	71.34	2.57	3,518	0.072

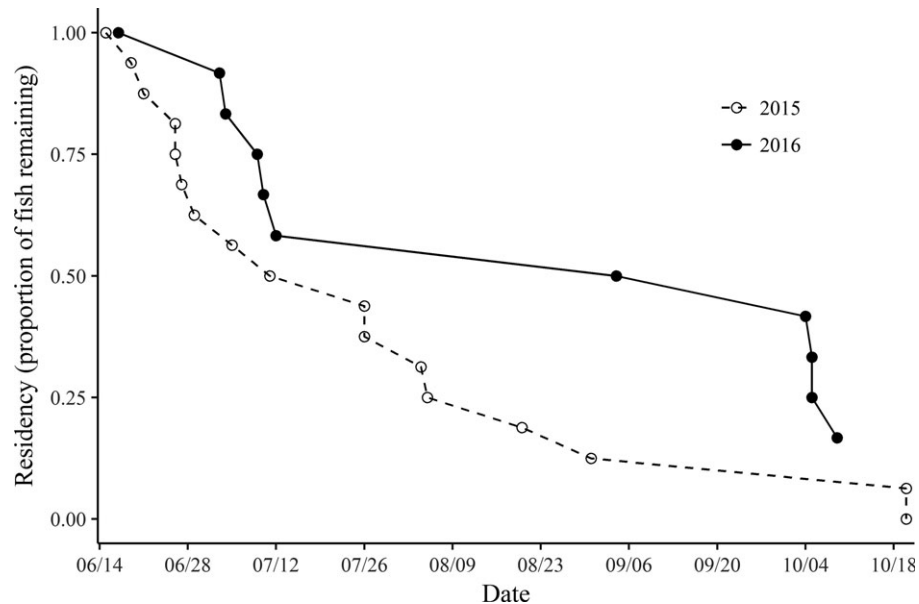


FIGURE 2. Residency of juvenile Sablefish in St. John Baptist Bay, Alaska, from mid-June to mid-October of each study year. Lines show the proportion of fish remaining within the receiver array over time.

head, and were categorized as mouth residents. Six fish were consistently detected closer to receiver 1, spending more time near the head of the bay but making short-term excursions away before returning; these fish were categorized as head residents. Twelve Sablefish moved periodically between the head and the mouth of the bay and were categorized as dispersive individuals. Of the mouth residents, all but two were tagged in 2016, while the head residents were split evenly across tagging years. All but one of the dispersive-type fish were tagged in 2015.

Visual classification of the individual plots was largely corroborated by the cluster analysis but differed for six fish. One fish (ID 43447) was visually classified as a mouth resident but was clustered as a head resident. Two

fish (IDs 43443 and 43451) were visually classified as mouth residents but were clustered into the dispersive group. The remaining three fish visually appeared to be dispersive (IDs 34757, 34772, and 43444) but were clustered with head residents. Overall, the analyses suggested that 25–36% of the fish were more closely associated with the mouth of the bay, 21–36% were more closely associated with the head of the bay, and 39–43% of the fish exhibited more dispersive movement patterns within the bay.

Body Size and Movement

Days present within the bay were significantly negatively related to fish FL (linear regression: $F = 10.69$,

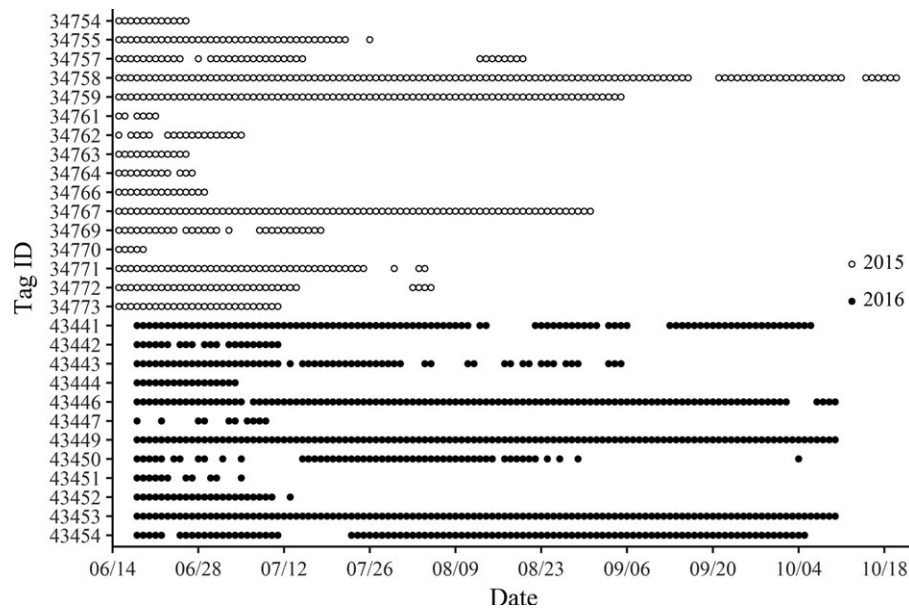


FIGURE 3. Calendar plot of juvenile Sablefish detections in St. John Baptist Bay, Alaska, by tag identification number (ID) from mid-June to mid-October of each study year. Each dot represents presence for each day on which a given fish was detected.

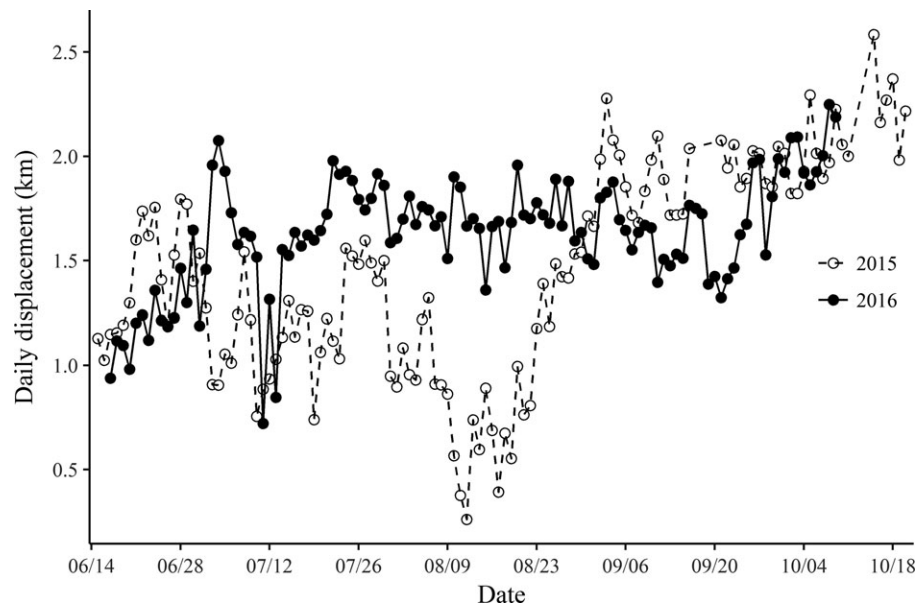


FIGURE 4. Plot comparing mean daily displacement (km) from receiver 1 at the head of St. John Baptist Bay, Alaska, over time for juvenile Sablefish tagged during each study year.

df = 26, $P = 0.003$). No relationship was found between FL and mean daily distance traveled within the bay (linear regression: $F = 0.14$, df = 26, $P = 0.71$). There was a significant negative linear relationship between FL and the mean daily duration (linear regression: $F = 11.50$, df = 26, $P = 0.002$). The coefficients of determination for both significant effects were moderate ($R^2 = 0.29$ and 0.31 , respectively).

DISCUSSION

Juvenile Sablefish spend the first 1–2 years of life in nearshore habitats before emigrating to offshore habitats, but relatively little is known about the extent of their residence and movement within these nearshore areas (Mason et al. 1983; Kendall and Matarese 1987; Rutecki and Varosi 1997a; Maloney and Sigler 2008). Previous telemetry research conducted on juvenile Sablefish focused on

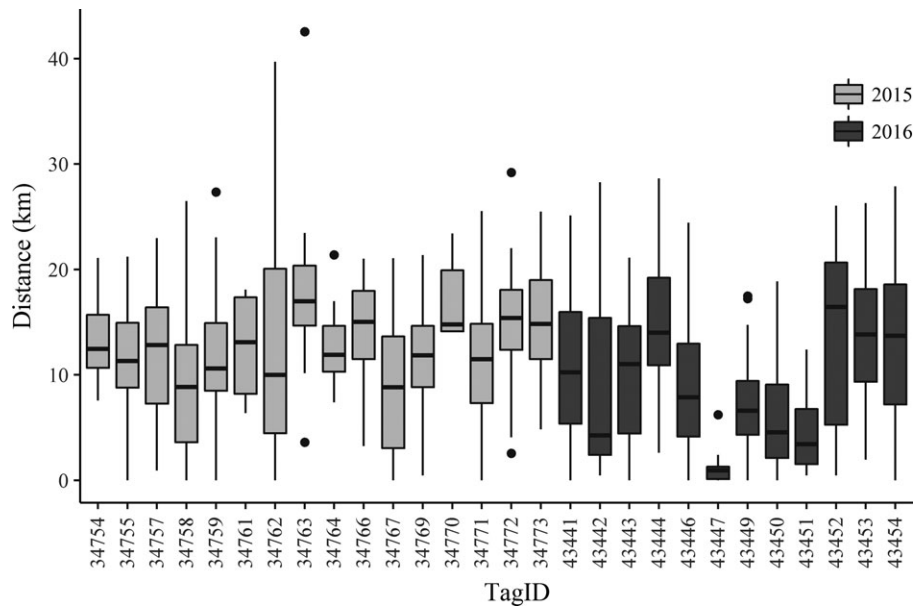


FIGURE 5. Daily distance traveled in St. John Baptist Bay, Alaska, by tagged juvenile Sablefish for all days detected during each study year (tagID = tag identification number). The black line within each box represents the median daily distance traveled, while the lower and upper edges of the box correspond to the first and third quartiles. The upper whisker extends to the greatest distance traveled no larger than 1.5 times the interquartile range, and the lower whisker extends to the smallest distance traveled at most 1.5 times the interquartile range. Dots beyond the whiskers represent outlier points.

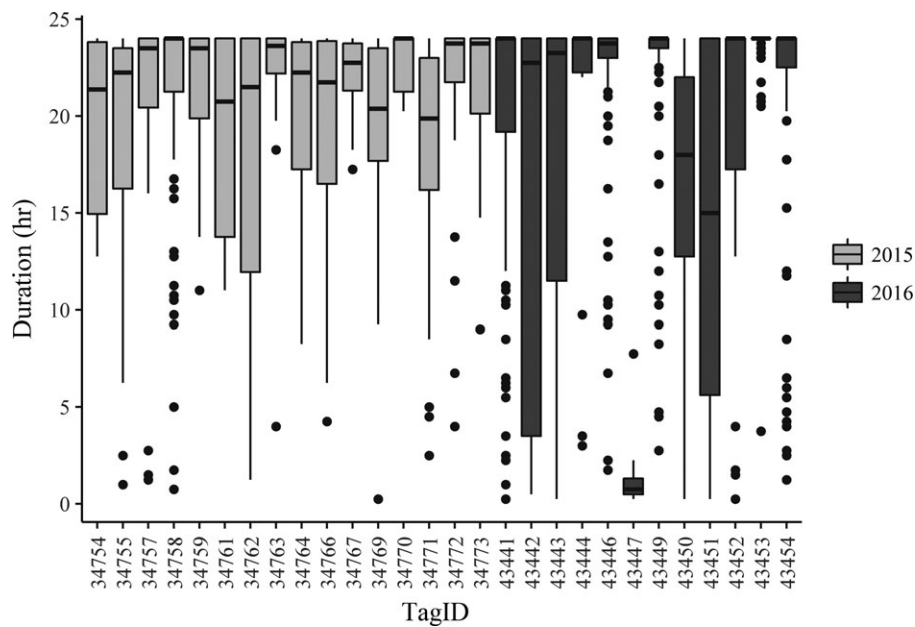


FIGURE 6. Daily duration spent by tagged juvenile Sablefish within the receiver array in St. John Baptist Bay, Alaska, for all days detected during each study year (tagID = tag identification number). The black line within boxes represents the median daily duration, while the lower and upper edges of the box correspond to the first and third quartiles. The upper whisker extends to the greatest duration no larger than 1.5 times the interquartile range, and the lower whisker extends to the smallest duration no further than 1.5 times the interquartile range. Dots beyond the whiskers represent outlier points.

vertical movements in a small area of SJBB over 40 d (Coutré et al. 2017). Other tagging studies have examined movement rates of over 34,000 juveniles that were tagged

externally with plastic T-bar anchor tags and released in Southeast Alaska (most in SJBB) from 1985 to 2005, with recoveries throughout the Gulf of Alaska (Rutecki and

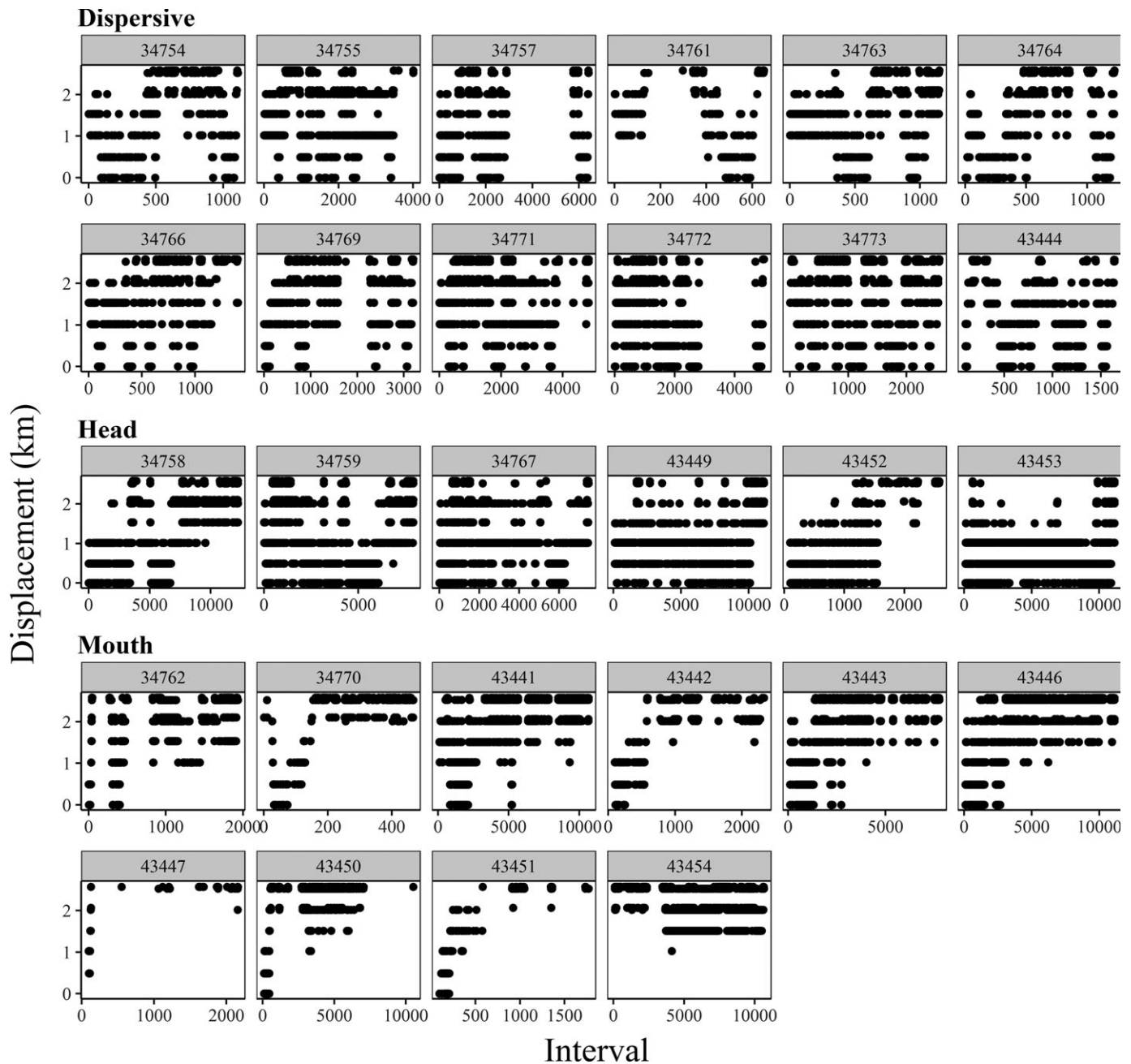


FIGURE 7. Movement plots for individual juvenile Sablefish, showing displacement distance from receiver 1 in St. John Baptist Bay, Alaska, over all 15-min intervals, organized by movement type (tag identification number is indicated at the top of each plot). Dispersive fish movement plots are shown in the upper two rows, head-resident movement plots are presented in the middle row, and mouth-resident movement plots are displayed in the lower two rows. Movement types shown are based on visual assessment. Visual assessments differed from cluster analysis results for six fish: 43443 and 43451 were assigned to the dispersive group, while 34757, 34772, 43444, and 43447 were identified as head residents based on cluster analysis.

Varosi 1997b; Maloney and Sigler 2008; Hanselman et al. 2015). Why SJBB remains a consistent location where juvenile Sablefish may be found is unknown, as an earlier study revealed no significant differences in temperature and salinity from conductivity–temperature–depth (CTD) measurements in SJBB relative to nearby bays (Rutecki and Varosi 1997a). This study fills a gap in knowledge of

Sablefish early life history by characterizing juvenile Sablefish movements during their summer and fall residency in SJBB, an important nursery area that has been a focus of ecological research since the 1980s (Rutecki and Varosi 1997a; Sigler et al. 2001; Maloney and Sigler 2008; Courtney and Rutecki 2011; Coutré et al. 2015, 2017).

Presence/Absence of Fish in the Bay

According to Rutecki and Varosi (1997a), juvenile Sablefish emigrate out of the nearshore habitats by the end of the summer, which could explain the progression of absences we observed over the course of both study years. Half of the fish tagged in 2015 were no longer detected by mid-July, while half of the fish tagged in 2016 were no longer detected by early September, leading us to conclude that they had left the bay for adjacent nearshore waters like Salisbury Sound, perhaps en route to offshore habitats in the Gulf of Alaska. It is not known what triggers juvenile Sablefish to leave nearshore habitats; however, by ages 3 and 4, most juvenile Sablefish are found offshore on the continental shelf at depths greater than 100 m (Maloney and Sigler 2008). No clear trend was observed marking an exodus from the nursery area; rather, individuals left the bay over the course of the season, with few fish remaining by October of either year. The progression of likely emigration was more gradual in 2016 compared to 2015; reasons for this difference were not explored in this study, but it may be due to differences in water temperatures, prey availability, predator densities, or Sablefish nutritional condition between years. Transmitters implanted into fish in 2015 were expected to last 370 d; however, after all fish appeared to have left the receiver array by October 2015, a single fish was detected again for a period of 4 d in December but was not detected after that. The 16 fish tagged in 2015 likely moved away from the receiver array within SJBB, although they could have experienced mortality or their tags could have malfunctioned. Transmitters inserted into fish in 2016 were expected to last approximately 476 d based on battery life, but the receivers were permanently removed from the bay in October 2016, although some fish were still detected daily in the weeks before the end of the study. Periods of time without detections in some fish may be related to the use of areas beyond detection range of receivers or movement outside of the bay entirely. Future telemetry studies could tag age-0 juvenile Sablefish in late fall to examine movement patterns and occupancy in SJBB throughout the entirety of their nursery inhabitation, including the overwinter period. With current technology, age-0 Sablefish may be too small to successfully carry acoustic tags that would have sufficient battery life to cover this time frame, and a previous tagging study found that age-0 Sablefish experienced a high rate of mortality when tagged in October compared to summer (Rutecki and Varosi 1997b).

Movement Patterns

We hypothesized that juvenile Sablefish would move throughout the entire bay but would show stronger fidelity to the head of the bay than the mouth. The highest RI values were not found at the very head of the bay at

receivers 1 and 2; instead, they were found in the middle head of the bay at receivers 2 and 3. This is corroborated by hook-and-line tagging surveys conducted by NMFS in SJBB (Dana Hanselman, NOAA, personal communication) as well as several ecological studies of juvenile Sablefish (Rutecki and Varosi 1997a; Courtney and Rutecki 2011; Coutr   et al. 2015, 2017; this study), which concentrated their sampling efforts in the middle-head area due to higher catch rates of juvenile Sablefish in that location. The affinity for the middle head of the bay in late summer could be due to the presence of postspawning salmon carcasses, which juvenile Sablefish have been documented to consume (Coutr   et al. 2015). Displacement of Sablefish toward the head of SJBB from August 8 to 23, 2015, corresponded to the time in which large numbers of Pink Salmon *O. gorbuscha* were entering the freshwater system at the head of the bay based on observations during mobile tracking surveys and Alaska Department of Fish and Game (ADFG) aerial surveys (Aaron Dupuis, ADFG, personal communication).

The middle-head area of the bay also could be more frequently occupied than other areas of the bay due to the habitat structure; for example, the southern shoreline near the mouth of the bay was notably rockier and steeper and produced the lowest RI values (receivers 6 and 8). Pirtle et al. (in press) found that juvenile Sablefish habitat was predicted to occur in low-lying areas with little rocky substrate and a low-gradient slope, including deep main channels of bays. The reference tag detection rate was similar between the two receivers at the mouth (receivers 7 and 8) and between the two at the middle mouth (receivers 5 and 6), suggesting that differences in occupancy for those areas were more likely due to tagged fish traveling along the northern shoreline than a result of acoustic shadowing by rocky habitat along the southern shoreline. Though not examined in this study, Sablefish may move to navigate tradeoffs in their thermal experience, food availability, and the degree of predation risk in nursery habitats. For example, Sogard and Olla (1998) found that juvenile Sablefish displayed avoidance of cold water in thermally stratified laboratory tanks but made potentially lethal vertical movements below the thermocline when food was present.

Variation in Individual Movements

Our results supported the hypothesis that individual Sablefish varied in their movement patterns and location within the bay. Displacement, distance, and duration were wide ranging among individuals and varied between years. For the 28 fish that were used in analyses, there was individual variation in their horizontal movements, measured by displacement from receiver 1 over time. Individual variation in movement types has been documented for other fishes, including Lingcod *Ophiodon elongatus*

(Beaudreau and Essington 2011), Pacific Halibut *Hippoglossus stenolepis* (Nielsen et al. 2014), wild juvenile Atlantic Salmon *Salmo salar* (Roy et al. 2013), coral reef fishes (Schoolmaster *Lutjanus apodus* and Stoplight Parrotfish *Sparisoma viride*; Garcia et al. 2015), and White Croaker *Genyonemus lineatus* (Wolfe and Lowe 2015). Sources of variation in movement types among individuals could be related to ecological factors, such as competition among conspecifics for territories, foraging strategies, predator avoidance, seasonal migrations, and physiological responses to the environment (Boesch and Turner 1984; Roy et al. 2013; Finn et al. 2014; Henderson et al. 2014; Sheaves et al. 2015). Heterogeneity in movement patterns at a microhabitat scale illustrates the dynamic nature of foraging decisions, mortality risks, competition, and life history traits like sex-specific behaviors and movements (Roy et al. 2013; Garcia et al. 2015). Our tagging data may be used in combination with environmental data to identify potential mechanisms that contribute to variation in movement patterns among juvenile Sablefish.

We expected that larger individuals that are closer to out-migration would show higher movement rates and lower residence in the bay. A marked ontogenetic shift in displacement, distance, or duration was not obvious in the current study. Significant negative relationships were found between fish FL and days present within the bay, as well as between FL and daily duration, but the small sample size and limited size range of juveniles tagged were not sufficient to allow for a strong conclusion regarding the effects of length. Nevertheless, these relationships may explain some of the differences we observed between years, specifically the progression of emigration, as fish tagged in 2015 were slightly larger than those tagged in 2016 (Table 1). Although displacement, distance, and duration within the bay varied between years and among individuals, both study years exhibited a gradual trend of increasing displacement distance from the head over the late summer and fall as tagged individuals out-migrated from the bay.

Passive acoustic telemetry allows researchers to study long-term movement monitoring, although not without caveats. Some disadvantages of using acoustic telemetry include the invasive nature of transmitter implantation, which may adversely affect the fish and its behavior (Eiler and Bishop 2016). Although it is sometimes apparent when a tag is expelled or mortality occurs, one may not be able to tell whether a tagged fish was consumed by a predator; however, methods to identify probable predation events by using predation tags are in development (Halfyard et al. 2017). For the 12 tagged fish that were excluded from analyses, we inferred that the tags were expelled or the fish did not survive the tagging procedure because the location of the tags ceased to change for over 30 d. Additionally, receiver orientation, placement of receivers in relation to

habitat structure, and depth of receivers affect the detection range and detectability of the transmitters (Huvener et al. 2016). Range testing can provide insight on potential placement issues; however, the risk of entanglement with fishing gear or anchor lines is also of concern. At least three receivers over the course of the study were encountered and had groundlines cut by fishing gear, but all receivers either were returned to researchers or were eventually recovered via grappling. Cost of acoustic telemetry, confinements of the study area, and limited battery life of the transmitters may further restrict study time and sample sizes (Eiler and Bishop 2016). For this study, the small size of the bay and the expected level of acoustic interference from vessel traffic restricted our sample size to 20 fish in each study year (Jonathan Mulock, Vemco, personal communication).

Adult Sablefish that were tagged as juveniles in coastal bays of Southeast Alaska, namely in SJBB, have been recovered as far west as the Aleutian Islands, as far north as the Bering Sea, as far south as off the coast of Vancouver Island, and even inland in the Chatham Strait fishery, illustrating that adult Sablefish are capable of considerable movement (Rutecki and Varosi 1997b; Maloney and Sigler 2008). We found that juvenile Sablefish traveled 9.4 km/d on average in 2015 and 13.0 km/d on average in 2016, suggesting that they are making substantial movements early in their lives, even in a small near-shore bay. Sablefish populations depend on recruitment from strong year-classes, and widespread abundance of age-1 juvenile Sablefish is indicative of a strong year-class (Rutecki and Varosi 1997a). Knowing the extent to which juvenile fish use nursery habitat and describing the ranges of movements during their residence are critical for defining juvenile life history, establishing habitat suitability models, and understanding the importance of nearshore nursery habitat in Southeast Alaska to recruitment success in both federal and state-managed fisheries of the North Pacific.

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